REPORT BRIEF — Autonomous Construction Equipment: Who Actually Owns the IP?

Report type: Technical Patent Landscape Analysis Report **Mapped GreyB service:** Patent Landscape Analysis + FTO Analysis + White Space Identification **Estimated length:** 2,500–3,500 words + visuals **Primary audience:** CTO, Chief IP Counsel, VP Engineering at Caterpillar, Komatsu, Volvo CE, John Deere, XCMG, Built Robotics

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THE TECHNICAL PROBLEM — FOR THE TEAM TO UNDERSTAND FIRST

Autonomous construction equipment is not one invention. It is a **stack of 8 distinct technology layers** — and each layer has its own patent landscape, its own dominant filers, and its own degree of white space still available.

Most people — including most clients — think of autonomy as a single thing. "The machine drives itself." But from a patent perspective, "the machine drives itself" requires solving 8 separate engineering problems, each of which can be independently patented, licensed, blocked, or designed around.

This is the stack the team needs to understand before writing anything:

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LAYER 1 — PERCEPTION
What it does: The machine must see its environment
Technologies: LiDAR, radar, stereo cameras, ultrasonic sensors,
         sensor fusion algorithms
Patent question: Who owns the sensor fusion algorithms
           optimised for outdoor unstructured terrain
           (mud, dust, variable light)?
Key distinction from automotive: Automotive autonomy
           assumes roads. Construction assumes no roads.
           Different terrain = different patents.

LAYER 2 — LOCALISATION
What it does: The machine must know exactly where it is
Technologies: GPS, RTK-GPS, SLAM (Simultaneous Localisation
         and Mapping), IMU (inertial measurement units)
Patent question: Who owns GPS-denied localisation?
           (underground tunnels, under bridge structures,
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dense urban sites — GPS doesn't work)
Key distinction: This is where construction autonomy diverges hardest from automotive and drone IP.

LAYER 3 — PATH PLANNING
What it does: The machine must decide how to move
Technologies: Dynamic obstacle avoidance algorithms, terrain traversability analysis, real-time replanning when conditions change
Patent question: Who owns path planning for earthmoving specifically — where the machine is both navigating AND reshaping the terrain it's on?

LAYER 4 — GRADE CONTROL
What it does: The machine must achieve precise cut/fill targets
Technologies: Machine learning models trained on GPS + IMU + blade position data, sub-centimetre accuracy on blade/bucket movement
Patent question: This is Komatsu's strongest layer from its iMC system developed since 2013. What exactly do they own and how tightly does it cover competitor implementations?

LAYER 5 — MACHINE INTERFACE
What it does: The autonomy system connects to the machine's hydraulic and control systems
Technologies: CAN bus integration, proprietary OEM control interfaces, retrofit autonomy kit hardware
Patent question: This is where retrofit players (Built Robotics, Teleo) vs. native OEM designs split. Retrofit kits need to interface with OEM machines without infringing OEM interface patents.

LAYER 6 — FLEET COORDINATION
What it does: Multiple autonomous machines must work together without colliding or creating deadlocks
Technologies: Multi-agent path planning, task allocation algorithms, collision avoidance at fleet level, digital dispatching systems
Patent question: XCMG and SANY have filed heavily here since 2020. This is the layer Western OEMs have left most open — and where Chinese OEMs are most aggressively building positions.

LAYER 7 — DIGITAL JOBSITE INTEGRATION
What it does: Machine data connects to BIM models, project management software, and ERP systems
Technologies: APIs, data standardisation (IFC, LandXML), real-time telemetry, cloud data pipelines
Patent question: Trimble and Topcon hold surveying and machine control integration patents that touch this layer. How do autonomy players navigate these?

LAYER 8 — REMOTE OPERATIONS
What it does: A human operator can monitor and intervene from a remote location
Technologies: Low-latency video, haptic feedback, 5G/edge compute, operator interface software
Patent question: Teleo (Caterpillar investment), Phantom Auto, and others hold remote ops IP. Regulatory requirements in most markets require a human in the loop — so this layer is not optional.

WHAT THE REPORT ARGUES

Central argument: The companies winning the autonomous equipment race are not necessarily the ones with the best machines. They are the ones that understood the 8-layer IP stack and filed across it strategically. Komatsu filed at Layers 3 and 4 starting in 2013 — a 10-year head start that created a defensible position before competitors understood what was happening. XCMG is now doing the same at Layers 5 and 6 — the layers Western OEMs left open.

The finding the report should land on: There is still white space in this landscape — but it is concentrated in specific sub-layers, and it is closing. Companies that want to file in defensible white space have 12–18 months before the data from today's filings becomes public and competitors can see what territory has been taken.

EXACT REPORT STRUCTURE

Section 1 — Why Construction Autonomy Is Not Automotive Autonomy (400 words)

The team's job here is to explain the technical distinction clearly. This matters because most readers assume the patent landscape for autonomous vehicles applies to autonomous equipment. It does not — and understanding why is what makes this report credible.

Key points to make:

- Automotive autonomy assumes structured environments: roads, lane markings, traffic signals, known obstacles. Construction sites have none of these.
- A Tesla's perception system trained on road data fails immediately in a quarry or demolition site — variable terrain, no fixed reference points, airborne dust degrading sensors, moving humans in close proximity to heavy machinery
- Construction equipment does not just navigate terrain — it reshapes it. An autonomous excavator must plan a path through ground that will not exist in the same form when the task is complete. This is a fundamentally different planning problem.
- GPS-denied localisation is a near-constant condition on complex construction sites. Automotive GPS-denied patents (tunnel patents, parking garage patents) cover different scenarios than construction-specific GPS denial.
- This means the construction autonomy patent landscape is **distinct from automotive, mining, and agricultural autonomy** even where the technologies appear similar. A Waymo patent covering road-based obstacle avoidance does not protect a Komatsu excavator from a competing implementation in an earthmoving context.

Why this matters for the report: It establishes that a dedicated construction-specific landscape study is necessary — you cannot just read across from automotive IP analysis.

Section 2 — The 8-Layer Stack: Where IP Concentration Is Dense vs. Open (700 words)

This is the core technical section. The team should treat each layer as a separate sub-section with three elements:

1. What the technology does (2–3 sentences, plain language)
2. Who the dominant filers are (name companies and approximate filing volume)
3. White space assessment (open / closing / closed)

Research guidance for the team — what to look for per layer:

Layer 1 (Perception): Look for: sensor fusion patents specifically citing "unstructured outdoor terrain" or "construction site" or "earthmoving" in claims. Key filers to check: Caterpillar, Komatsu, Velodyne (LiDAR), Ouster, Continental (which supplies sensors to OEMs). The construction-specific claim language is the differentiator — automotive sensor fusion patents use road-specific claim language.

Layer 2 (Localisation): Look for: SLAM patents combined with heavy machinery, RTK-GPS in construction contexts, inertial navigation for large vehicles. Key filers: Trimble (strongest position here from machine control heritage), Topcon, Leica Geosystems (Hexagon). Also check Robust.AI and Exodigo (underground localisation startups).

Layer 3 (Path Planning): Look for: patents claiming path planning for "earthmoving," "terrain deformation," or "variable ground surface." Key filers: Komatsu (strongest position from iMC), Carnegie Mellon University (foundational robotics path planning, licensed broadly), Caterpillar. Note: many foundational path planning patents are now expired — the white space question is about construction-specific implementations.

Layer 4 (Grade Control): Look for: Komatsu iMC patent family — this is a specific cluster to map in full. Claims around blade/bucket positioning using machine learning, GPS + IMU fusion for sub-centimetre accuracy. Also check: Trimble (GCS900 grade control system), Leica Geosystems, Case Construction (CNH Industrial).

Layer 5 (Machine Interface): Look for: retrofit autonomy kit patents, CAN bus integration for hydraulic control, fly-by-wire conversion patents. Key filers: Built Robotics (Guident), Hilti (post-Canvas acquisition), Doosan (CAN bus integration patents). This layer is where startup IP sits closest to OEM proprietary interfaces — FTO risk is highest here.

Layer 6 (Fleet Coordination): Look for: multi-agent task allocation for construction equipment, collision avoidance between heterogeneous machines (excavator + dumper + compactor operating simultaneously), digital dispatching. Key filers: XCMG (flag high volume post-2020), SANY, Volvo CE (TARA autonomous transport system). This is the layer where Chinese OEM filing density is highest and Western OEM coverage is thinnest.

Layer 7 (Digital Jobsite Integration): Look for: machine telemetry to BIM, LandXML/IFC data integration with equipment control, digital twin connection to physical machine state. Key filers: Trimble (strongest from Viewpoint acquisition), Autodesk (from Pype and BuildingConnected acquisitions), Hilti (ON!Track integration). Note: this layer overlaps ConTech patent territory — OEM clients may not realise they are entering ConTech IP space.

Layer 8 (Remote Operations): Look for: low-latency video with haptic feedback for heavy machinery, 5G/edge compute for real-time remote control, operator interface systems for excavators specifically. Key filers: Teleo (backed by Caterpillar Ventures), Phantom Auto, Nuro (cross-domain). Regulatory note: most markets require remote human oversight capability — so this layer is not optional for commercial deployment, making the IP critical.

Section 3 — The Chinese Filing Pattern at Layer 6 (500 words)

This section makes the competitive intelligence argument specific and evidence-based. The team needs to show — not just claim — that XCMG and SANY have filed aggressively at fleet coordination.

What to research and show:

- Pull XCMG and SANY's patent filing volume in fleet coordination / multi-agent systems by year (2018, 2019, 2020, 2021, 2022, 2023, 2024) — the inflection point should be visible around 2020–2021
- Show jurisdictions: China first, then PCT, then specific jurisdictions (US, EU, South Korea, Japan) — this is the encirclement sequence
- Compare to Caterpillar and Komatsu filing volume in the same Layer 6 space over the same period
- The argument: Western OEMs concentrated their filing at Layers 3 and 4 (path planning, grade control) where they had existing expertise. They underinvested at Layer 6 (fleet coordination) because multi-machine autonomy was further from commercialisation. XCMG identified this gap and has been filling it.

Supporting data point to include: XCMG moved from #6 to #3 in global equipment sales (Yellow Table 2026). Revenue growth is funding an R&D and patent programme that is now filing at the rate of a technology company, not a traditional equipment OEM.

Section 4 — The FTO Problem Nobody Is Talking About (400 words)

This section introduces the practical problem that most contractors and equipment buyers have not considered.

The scenario to explain:

A major contractor is running a quarry or infrastructure project. They have:

- A Komatsu PC210 excavator with iMC (Layer 4 — Komatsu IP)
- A Built Robotics retrofit autonomy kit added to a Caterpillar 320 (Layer 5 — Built Robotics IP + Caterpillar machine interface)
- Trimble GCS900 grade control connected to both machines (Layers 2, 4, 7 — Trimble IP)
- A fleet coordination software from a startup (Layer 6 — potentially XCMG-adjacent IP space)
- All machines uploading to Autodesk Tandem digital twin (Layer 7 — Autodesk IP)

This contractor has assembled a system that spans IP from at least 5 separate ownership trees — Komatsu, Built Robotics, Caterpillar, Trimble, and Autodesk. Nobody in the transaction — not the contractor, not the OEMs, not the software vendors — has done a freedom-to-operate analysis on the combined system.

Why this is a real risk, not a theoretical one:

- NPEs (non-practising entities / patent trolls) specifically target technology integrations where multiple IP streams converge — because no single entity owns the whole stack and therefore no single entity has done the combined FTO
- The $6.57B in construction technology investment in 2025 has created dozens of well-funded startups that are now scaling into enterprise contracts — the moment a startup signs a Fortune 500 construction company as a client, it becomes an NPE target

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Section 5 — White Space: Where There Is Still Room to File (400 words)

This is the actionable section — what companies can actually do with this information.

White space areas the team should identify:

- **GPS-denied localisation for large equipment in urban construction** — automotive GPS-denied patents cover tunnels and parking; construction-specific claims in dense urban environments (surrounded by tall buildings, under bridges, near infrastructure) are underserved
- **Terrain deformation path planning** — planning algorithms that account for the fact that the ground being navigated is also being modified by the machine doing the navigating; foundational patents exist but construction-specific implementations are thin
- **Heterogeneous fleet coordination** — most fleet coordination patents cover homogeneous fleets (all excavators, all dump trucks); coordination between different machine types working on the same task is underpatented
- **Remote operations for compliance** — regulatory requirements for human oversight are now being codified in multiple jurisdictions; the specific technical implementations that satisfy these requirements are underpatented because the regulations are new

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Section 6 — What Companies Should Do (300 words)

Keep this practical. Three clear actions:

Action 1 — Run a layer-by-layer patent landscape before committing R&D capex Before a company invests in developing any one of these 8 layers, they need a landscape that tells them: who owns what, what the claim scope actually covers, and where they can file without conflict. This is not optional due diligence — it is the difference between a 20-year defensible IP position and a development programme that ends in a licensing demand.

Action 2 — Do a combined FTO analysis on any integrated system If a company is combining their own technology with third-party hardware or software across multiple layers, a combined FTO analysis is required. The risk is not any single component — it is the interaction between components from different IP owners.

Action 3 — File in white space now, not when the landscape is fully visible The 18-month publication lag means the white space that exists today in public patent data is already being occupied by filings that will not become visible until late 2026 or 2027. Companies that act on white space analysis now will have filed before competitors can see the territory was taken. Companies that wait for the data to become fully visible will have missed the window.

DATA SOURCES FOR THE TEAM

Source	What to use it for
Google Patents / Espacenet	Initial landscape — search by assignee (Komatsu, XCMG, Caterpillar, Built Robotics) + IPC class (G05D for autonomous systems, E02F for earthmoving equipment)
USPTO Patent Full-Text Database	US filing detail — claim language analysis
WIPO PatentScope	PCT applications — see international filing sequences
Lens.org	Free full-text search — good for citation network analysis
PatSnap / Derwent Innovation	If the team has access — pre-built landscaping tools
Yellow Table 2026	Equipment market share data — XCMG market position
Built Robotics press releases	Layer 5 IP context — their own description of their technology
Komatsu iMC product documentation	Layer 3/4 technical context
Trimble GCS900 technical specifications	Layer 2/7 context
McKinsey Global Infrastructure data	Market size and productivity context for the intro

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KEY TERMS THE TEAM MUST UNDERSTAND BEFORE WRITING

Term	Plain meaning
IPC Class	The international patent classification system — E02F = excavating/dredging machinery; G05D = systems for controlling position/direction; B60W = combined vehicle control — these are the search classes for this landscape
Claim scope	What a patent actually covers — not the headline, but the specific legal language in the independent claims. Two patents can look similar from abstracts but have completely different actual coverage.
Continuation filing	A new patent application filed after a parent application, using the same priority date — used to extend coverage to new implementations as the technology evolves. XCMG's fleet coordination filing pattern should be checked for continuation families.
PCT application	Patent Cooperation Treaty — a single international filing that reserves the right to pursue patents in 150+ countries. Filing a PCT before entering specific national phases is how encirclement is built — it secures the date without immediately incurring per-country costs.
FTO (Freedom to Operate)	An analysis of whether a specific product or process can be commercialised without infringing active patents in a given market.
White space	Technology areas that are patentable but currently unpatented — either nobody has thought of claiming it, or the market wasn't ready when earlier filings were made.
NPE / Patent troll	A company that owns patents but does not make products — exists solely to licence or litigate patents against companies that do make products. They specifically target technology integrations where multiple IP streams converge.
Prior art	Any publicly available evidence (patents, papers, products) that predates a patent filing and can be used to challenge its validity.

VISUAL CHECKLIST FOR THE REPORT

- [ ] **The 8-layer stack diagram** — vertical stack showing all 8 layers with colour coding: green = white space, amber = closing, red = densely occupied

- [ ] **Filing volume by year chart** — XCMG vs. Komatsu vs. Caterpillar filing volume in Layer 6 (fleet coordination) from 2018–2024 — the inflection point tells the story
- [ ] **IP ownership map** — which companies own which layers (a simple matrix: companies on rows, layers on columns, cell = strong/moderate/thin/none)
- [ ] **Jurisdiction filing sequence** — show how a typical Chinese OEM encirclement unfolds: CN first → PCT → US/EU/JP/KR — one diagram that explains the pattern
- [ ] **The FTO scenario illustration** — show the contractor using components from 5 IP owners simultaneously, with arrows showing where the risk junctions are

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ONE-LINE SUMMARY FOR THE TEAM

> Autonomous construction equipment is an 8-layer technology problem, and the companies that will own the market are the ones that mapped the IP landscape layer by layer, filed in the white space that still existed, and did FTO analysis before assembling multi-vendor systems — not after receiving a patent demand letter.